MiniProject1 - Text Analysis

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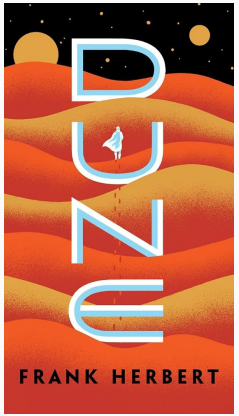
Research Question

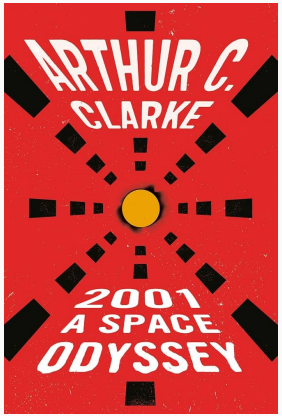
Are there any significant vocabulary differences between hard and soft science fiction, despite being the same genre?

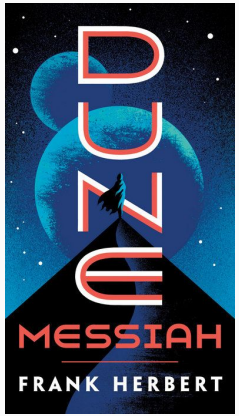


Goals

- Compare two works by Frank Herbert in his *Dune* series
- Compare Herbert's works with Arthur C. Clarke's *2001: A Space Odyssey*
- Observe vocabulary differences
  - Fantastical verbiage vs. technical jargon

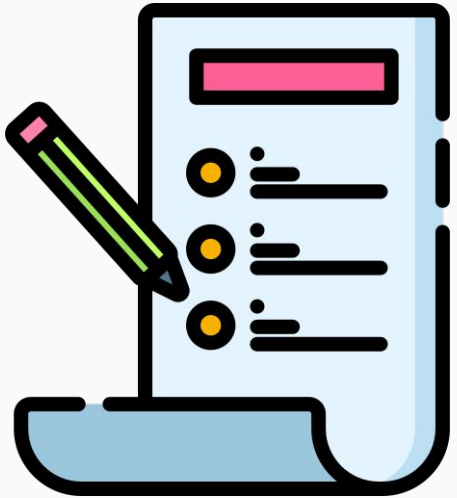






Approach

1. PDF Text Extraction and Cleaning
2. Plot Most Frequent Words (Normalized per 10,000 Words)
3. Cosine Similarity Between Books



PDF Text Extraction and Cleaning

- Stopwords included top 100 most common English words + NLTK's stopword list
- Extract text from book PDFs
- Strip all non-alphanumeric characters and number-only words
- Obtain top 15 frequent words per 10,000 regular words

```
# Concatenates the text of every page in a PDF into one string.
def extract_pdf_text(path):
    reader = PdfReader(path)
    return "\n".join(page.extract_text() or "" for page in reader.pages)

# Strips all non-alphanumeric characters and words that are only numbers.
def cleanWord(w):
    wn = re.sub('[^a-z0-9]', '', w)
    return re.sub('^[0-9]*$', "", wn)

# Returns the n most frequent words in the text, sorted ascending by frequency per 10,000
# regular (non-stopword) words, so books of very different lengths (e.g. Dune vs. Dune: Messiah)
# can be compared on equal footing.
def get_wf(text, n=15):
    wds = re.split('\s+', text.lower())
    wf = Counter(cleanWord(w) for w in wds)
    wf.pop('', None)
    for k in stop_words:
        wf.pop(k, None)
    tw = sum(wf.values())
    wfs = sorted(wf.items(), key=operator.itemgetter(1), reverse=True)
    ml = min(len(wfs), n)
    top = wfs[0:ml][::-1]
    normalized = [(word, count / tw * 10000) for word, count in top]
    return normalized, tw
```

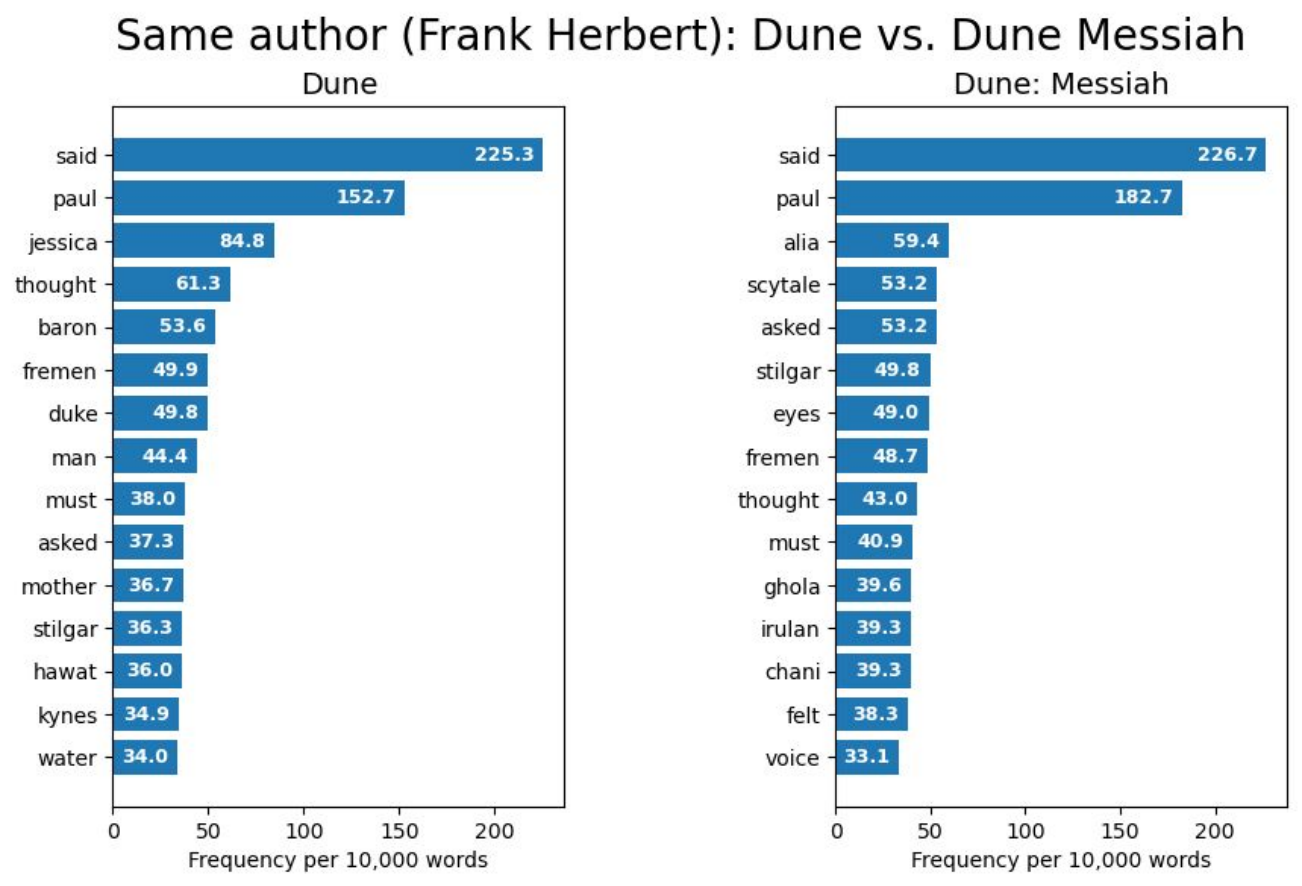
Total # of Regular Words + Word Frequencies per 10,000 Words per Book

```
--------------------------------------------------
Dune (100819 regular words)
--------------------------------------------------
Top 15 words (frequency per 10,000 words):
  water: 34.02
  kynes: 34.91
  hawat: 36.01
  stilgar: 36.30
  mother: 36.70
  asked: 37.29
  must: 37.99
  man: 44.44
  duke: 49.79
  fremen: 49.89
  baron: 53.56
  thought: 61.30
  jessica: 84.81
  paul: 152.75
  said: 225.26
```

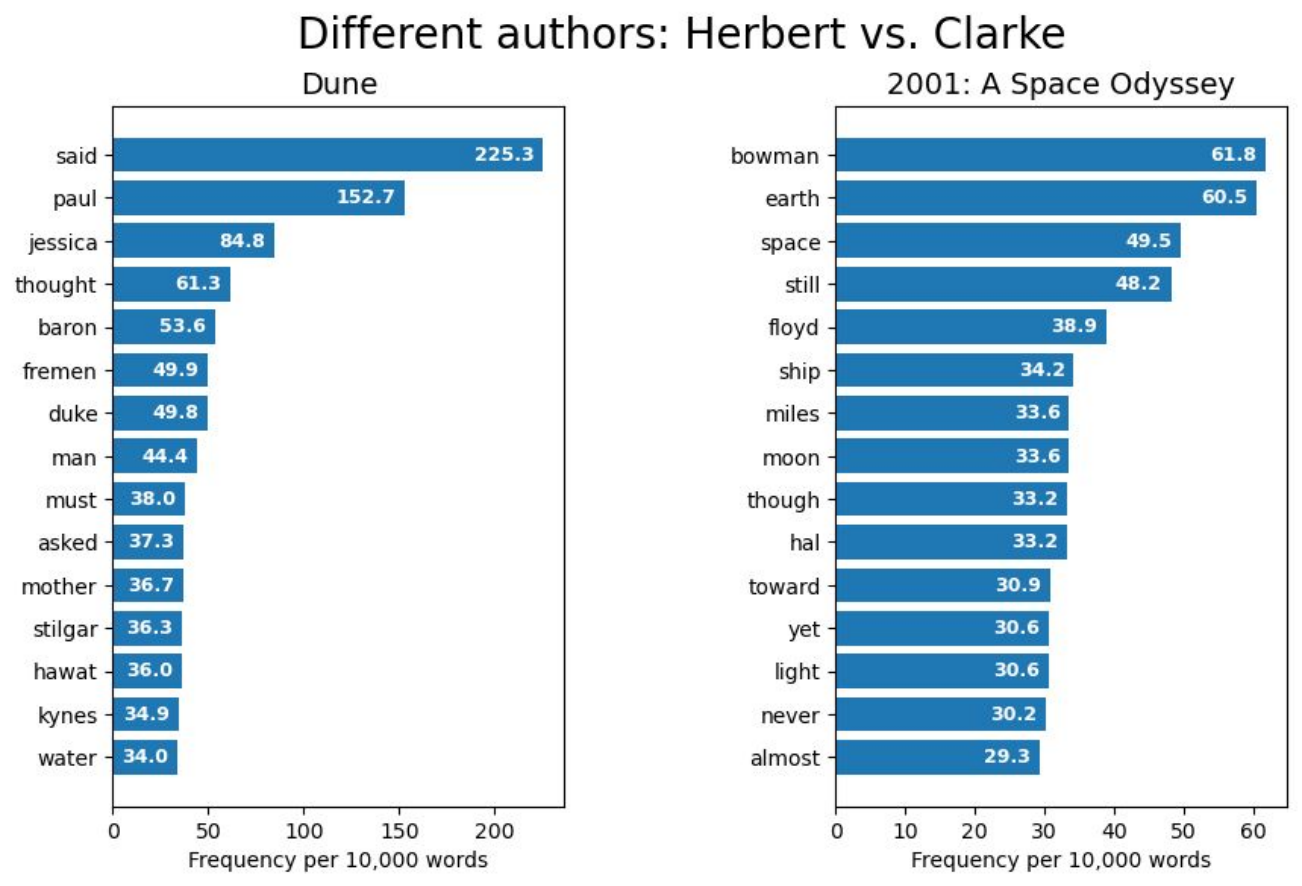
```
--------------------------------------------------
Dune: Messiah (38378 regular words)
--------------------------------------------------
Top 15 words (frequency per 10,000 words):
  voice: 33.09
  felt: 38.30
  chani: 39.35
  irulan: 39.35
  ghola: 39.61
  must: 40.91
  thought: 42.99
  fremen: 48.73
  eyes: 48.99
  stilgar: 49.77
  asked: 53.16
  scytale: 53.16
  alia: 59.41
  paul: 182.66
  said: 226.69
```

```
--------------------------------------------------
2001: A Space Odyssey (30084 regular words)
--------------------------------------------------
Top 15 words (frequency per 10,000 words):
  almost: 29.25
  never: 30.25
  light: 30.58
  yet: 30.58
  toward: 30.91
  hal: 33.24
  though: 33.24
  moon: 33.57
  miles: 33.57
  ship: 34.24
  floyd: 38.89
  still: 48.20
  space: 49.53
  earth: 60.50
  bowman: 61.83
```

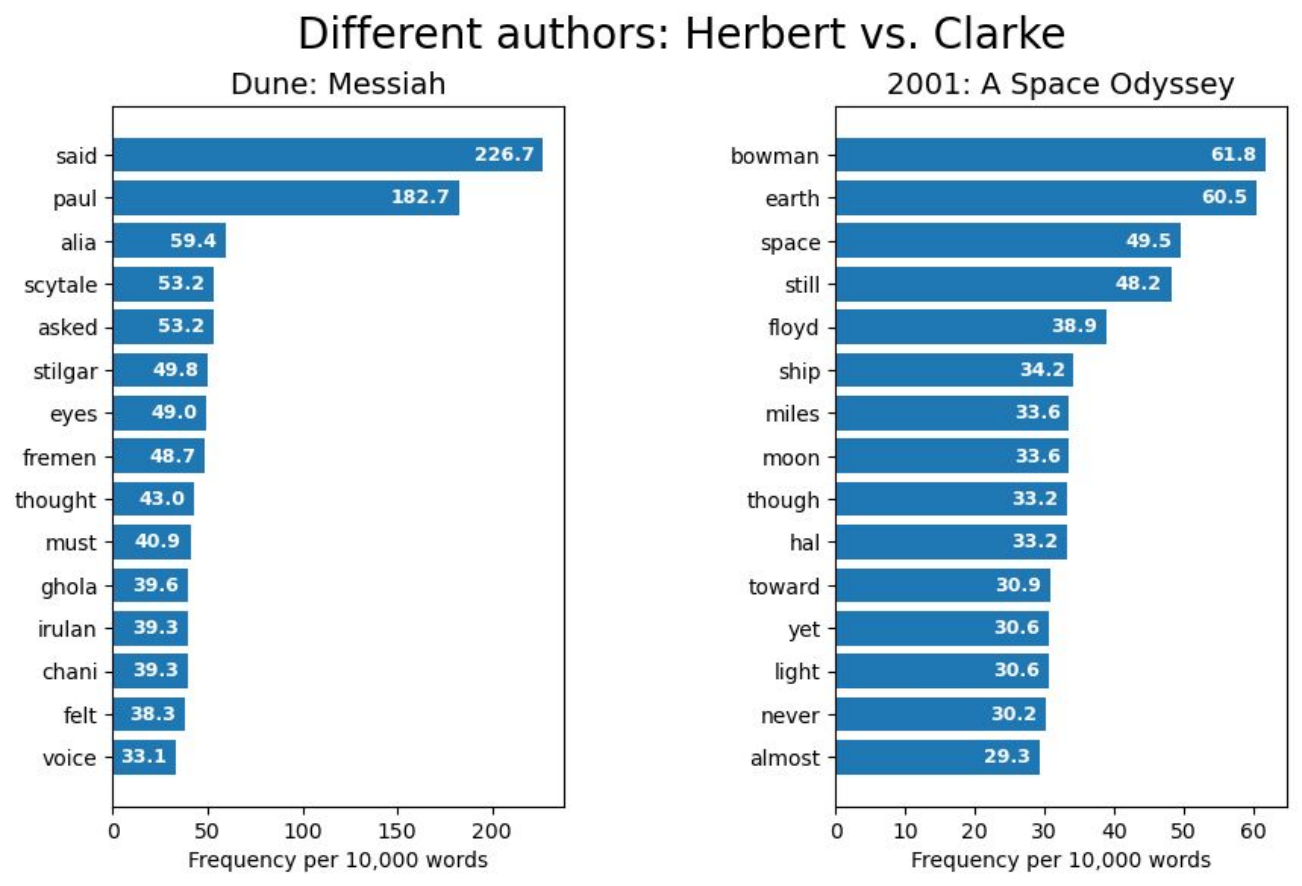
Plotting Most Frequent Words: *Dune* vs. *Dune: Messiah*



Plotting Most Frequent Words: *Dune* vs. *2001: A Space Odyssey*



Plotting Most Frequent Words: *Dune: Messiah* vs. *2001: A Space Odyssey*



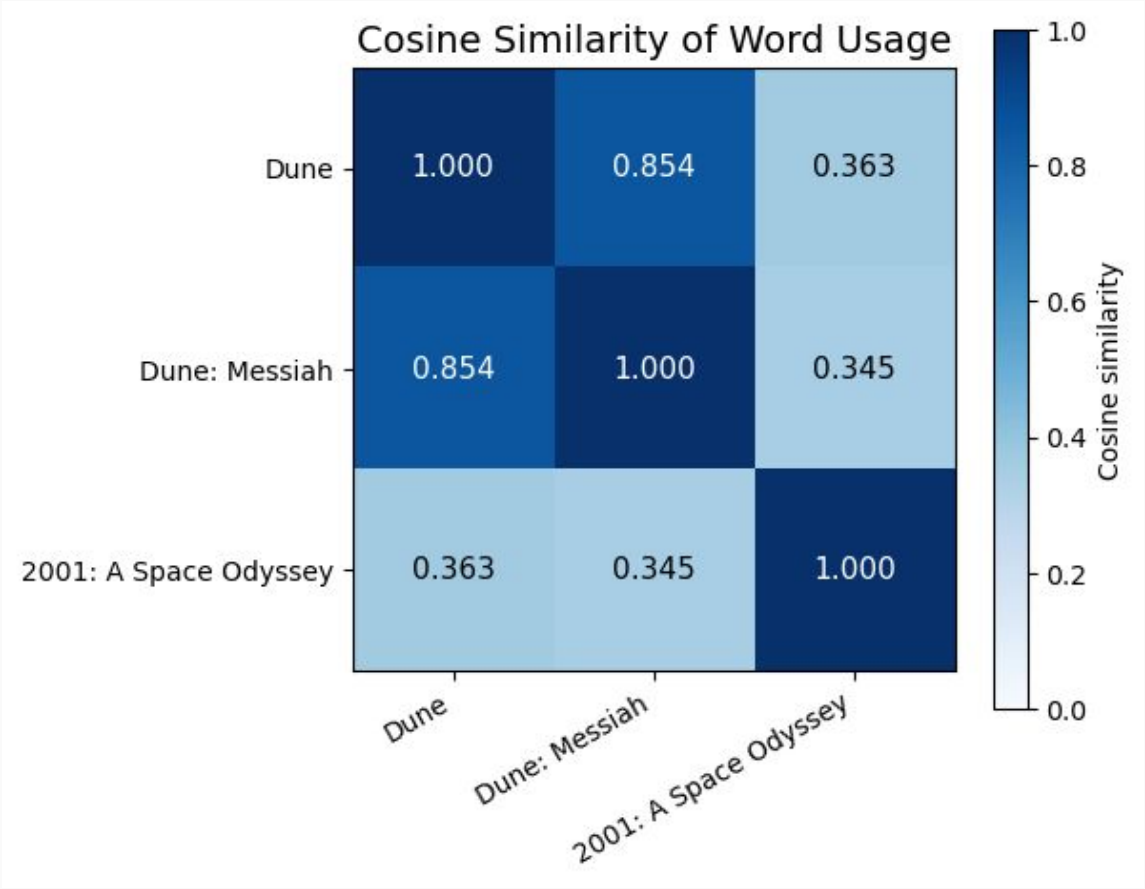
Cosine Similarity Between Books

- Top-word lists barely overlap; cosine similarity might provide better insights
- Used full stopword-filtered word-frequency counts for each book (not just top-n)

```
# Full (non-truncated) stopword-filtered word-frequency counts for each book, reusing the
# same cleaning as get_wf but keeping every word instead of just the top n.
def full_word_freq(text):
    wds = re.split('\s+', text.lower())
    wf = Counter(cleanWord(w) for w in wds)
    wf.pop('', None)
    for k in stop_words:
        wf.pop(k, None)
    return wf

full_freqs = {title: full_word_freq(text) for title, text in texts.items()}
```

Cosine Similarity Heatmap



Observations

1. Cosine similarity between the *Dune* books (**0.854**) is more than double the similarity between either *Dune* book and *2001* (**0.36**/**0.35**)
2. *2001*'s top words are more aligned with physical/spatial descriptions (space, earth, moon, miles, ship) while *Dune*'s are towards political/social nouns (duke, baron, mother) and *Messiah*'s are towards introspective words (voice, felt, eyes)
3. "said" and "Paul" in both *Dune* books have a near-identical normalized rate ("said": 225.26 vs 226.69; "paul": 152.75 vs 182.66)

Problems/Limitations

- Not available on project Gutenberg; needed to find PDFs with extractable text overlay
- NLTK's stopwords weren't sufficient on their own
- Only used three books



Conclusion + Future Ideas

- Project supports the idea that there are differences between hard and soft science fiction texts
  - Former focuses more on physical descriptions and science, latter focuses more on characters and narrative
- Include various texts from different authors in both subgenres to widen scope
- Inspect bigrams/trigrams moving forward